

LegalEase AI – Phase 3: Project Design Phase

1. INTRODUCTION

The **Project Design Phase** represents the third key phase in the development lifecycle of **LegalEase AI**. Following the successful identification and validation of functional and technical requirements during Phase 2, this phase establishes the structural blueprints and system workflows.

During this phase, the system architecture, client-side frontend interfaces, backend application services, AI/LLM modules, and internal data communication channels are formally designed. The fundamental goal of the design phase is to map out a clear data journey—from user query ingestion, through backend orchestrations and AI processing, to final response rendering—ensuring a scalable, user-centric, and well-structured legal assistance application.

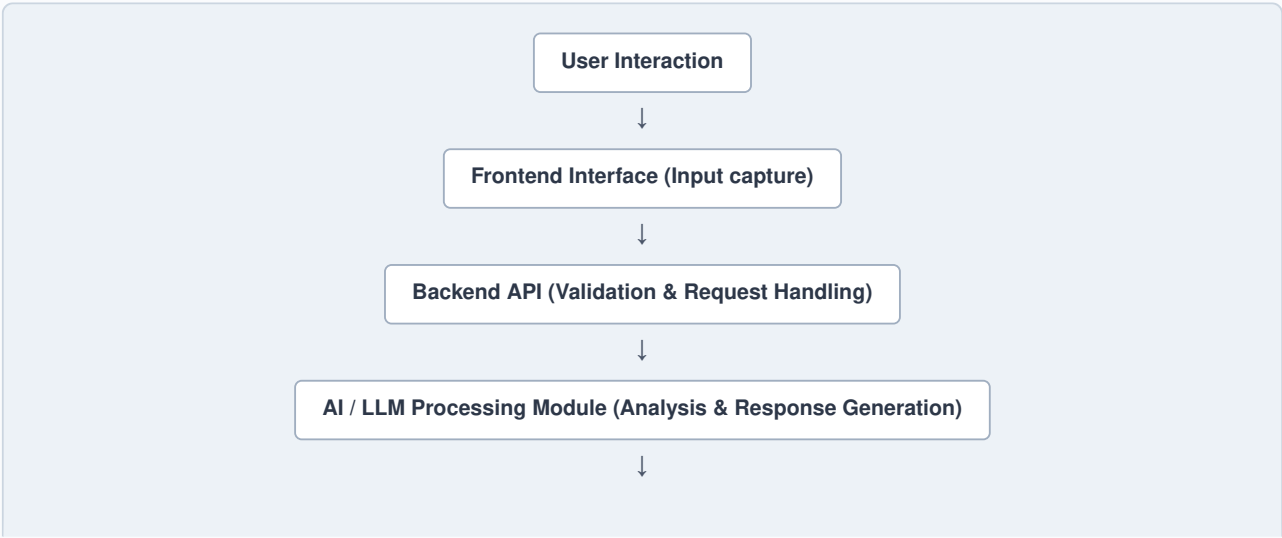
2. OBJECTIVES

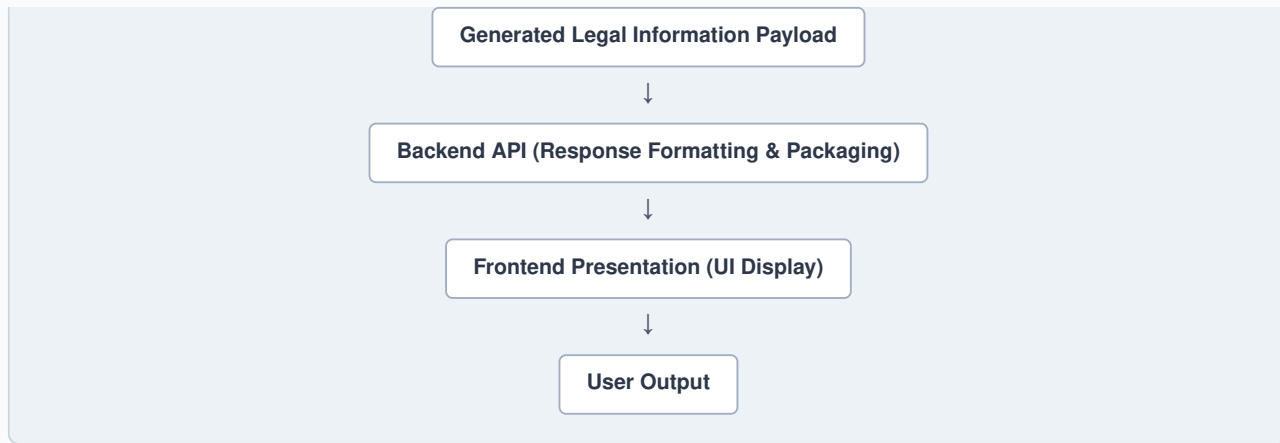
The core objectives of the Project Design Phase are:

- **System Architecture:** Design a robust, decoupled architecture connecting presentation, business logic, and AI layers.
- **Frontend Layout Design:** Plan a simple, responsive, and intuitive user interface tailored for seamless interaction.
- **Backend API Specification:** Design RESTful backend endpoints for input handling, payload validation, and task routing.
- **AI Module Integration:** Structure the AI/LLM orchestration layer for legal query answering and document processing.
- **Component Communication:** Map end-to-end data pipelines across frontend UI, backend Flask server, and external AI services.
- **Lifecycle & Workflow Definition:** Establish predictable application workflows and state management controls.

3. SYSTEM WORKFLOW

The system workflow follows a clear bidirectionally routed execution loop, tracing the life cycle of a query:





4. DETAILED DESIGN

4.1 Frontend Design

The frontend acts as the primary visual interface, prioritizing simplicity and accessibility for users without legal or technical expertise. Key UI design components include:

- **Application Header:** Clear Branding title (`LegalEase AI`) and context banner.
- **Input Console:** Multi-line prompt field supporting query text and document inputs.
- **Action Controls:** Interactive Submit and Clear options with active status feedback.
- **Response Section:** Rendered display container formatted with structured, readable typography.

4.2 Backend Design

The backend serves as the centralized orchestration engine. Built on a Python framework, its primary functional responsibilities include:

- Ingesting POST requests from the client application.
- Sanitizing, validating, and formatting input prompts.
- Dispatching payloads to external AI / LLM integration endpoints.
- Handling errors, network timeouts, and raw model response parsing.
- Serving structured JSON responses back to the client interface.

4.3 AI Module Design

The AI processing module manages domain-specific tasks using Large Language Models (LLMs). It is structured to handle:

- **General Legal Answering:** Processing queries on statutes, terms, and legal rights.
- **Information Simplification:** Translating dense legalese into plain, structured text.
- **Document Parsing Support:** Extracting core insights and clauses from user-provided contracts.

5. BASIC DESIGN CODE

Below is the conceptual Python / Flask architectural design for the primary backend communication endpoint:

```
from flask import Flask, request, jsonify app = Flask(__name__) # API endpoint design for
processing questions @app.route("/ask", methods=["POST"]) def ask(): data = request.get_json()
question = data.get("question") if not question: return jsonify({ "error": "Please enter a
question" }), 400 # AI/LLM integration layer hook response = "AI response will be generated
here." return jsonify({ "question": question, "response": response }) if __name__ ==
 "__main__": app.run(debug=True)
```

Architecture Explanation:

The /ask API acts as the core entry point for client requests. The server validates that the payload contains a query, routes the input to the AI engine wrapper, processes the result, and formats a standardized JSON output containing both original query and simplified legal response.

6. EXPECTED OUTCOME

The primary deliverables resulting from the Project Design Phase include:

- Comprehensive system architecture topology blueprint.
- Defined wireframe concept for the frontend UI.
- Restful API route design and JSON payload definitions.
- AI/LLM processing and simplification workflow map.
- Unified communication protocol between frontend, backend, and external LLM services.

7. PHASE CONCLUSION

The **Project Design Phase** establishes a concrete technical framework for building LegalEase AI. By formalizing component interfaces, data flows, and backend API interactions, this phase minimizes architectural ambiguity and lays a reliable foundation for the subsequent Project Planning and Active Development Phases.